Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Course : B.Tech
Code : 101619

Subject: Soil Mechanics - I

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) The consistency of clayey soils can be described by which test?
(i) Proctor test (iii) Sieve analysis
(ii) Atterberg limits test (iv) Hydrometer analysis

(b) The term "plastic limit" refers to:
(i) Moisture content at which soil can flow like a liquid (iii) Moisture content at which soil can be rolled into threads
(ii) The moisture content at which the soil just begins to crumble (iv) The maximum dry density achievable

(c) What does the term "void ratio" in soil refer to?
(i) Ratio of volume of voids to volume of soil (iii) Ratio of water content to the volume of voids
(ii) Ratio of volume of solids to volume of voids (iv) Ratio of density to moisture content

(d) The liquid limit of soil is determined by:
(i) Pycnometer (iii) Triaxial apparatus
(ii) Sieve analysis (iv) Casagrande's apparatus

(e) Which term refers to the gradual reduction in volume of a soil mass due to loading?
(i) Constitution (iii) Consolidation
(ii) Compression (iv) Compaction

(f) In a saturated soil sample, the degree of saturation is:
(i) 0% (ii) 50% (iii) 90% (iv) 100%

(g) The "coefficient of permeability" is a measure of:
(i) The soil's ability to retain water (iii) The soil's porosity
(ii) The rate at which water can flow through a soil (iv) The density of the soil

(h) The "specific gravity" of soil solids is typically in the range:
(i) 1.0 - 1.5 (iii) 3.0 - 3.5
(ii) 2.0 - 2.7 (iv) 4.0 - 4.5

(i) The main purpose of sieve analysis is to:
(i) Determine the compaction characteristics (iii) Measure the distribution of particle sizes
(ii) Find the specific gravity of soil (iv) Measure the water content of the soil

(j) For a soil sample, if the dry density is higher, the void ratio will be:
(i) Higher (iii) Constant
(ii) Equal (iv) Lower

Q.2 (a) A partially saturated sample from a borrow pit has a natural moisture content of 15% and bulk unit weight of 1.9 g/cc. The specific gravity of solids is 2.70. Determine the degree of saturation and void ratio. What will be the unit weight of the sample on saturation? **[7]**

(b) A clay sample originally 25 mm thick and at a void ratio of 1.12, was subjected to a compressive load. After the clay sample was completely consolidated, its thickness was measured to be 24 mm. compute the final void ratio. **[7]**

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Q.3 (a) The values of liquid limit, plastic limit and shrinkage limit of a soil were reported as follows: [7]

W_L=60%, W_P = 30%, W_S = 20%

If a sample of this soil at liquid limit has a volume of 40 cc and its volume measured at shrinkage limit was 23.5 cc, determine the specific gravity of the solids. What is its shrinkage ratio?

(b) In a falling head permeability test on a sample 12.2 cm high and 44.41 cm^2 in cross-sectional area, the water level in a standpipe of 6.25 mm internal diameter dropped from a height of 75 cm to 24.7 cm in 15 minutes. Find the coefficient of permeability. [7]

Q.4 (a) A concentrated load of 40 kN acts on the surface of a soil. Determine the vertical stress increment at points directly beneath the load upto a depth of 10 m and draw a plot. Also plot the variation of vertical stress increment due to load on horizontal planes at depth of 1m, 2m and 3m upto a horizontal distance of 3 m on either side of centre. [7]

(b) The water table in a deposit of sand 8 m thick, is at a depth of 3m below the surface. Above the water table, the sand is saturated with capillary water. The bulk density of sand is 19.62 kN/m^3. Calculate the effective pressure at 1m, 3m and 8m below the surface. Hence plot the variation of total pressure, neutral pressure and effective pressure over the depth of 8m. [7]

Q.5 (a) A cohesive soil yields a maximum dry density of 1.8 g/cc at an OMC of 16% during a standard proctor test. If the value of G is 2.65, what is the degree of saturation? What is the maximum dry density it can further compacted to? [7]

(b) A compressible layer is expected to have total settlement of 15 cm under a given loading. It settles by 3 cm at the end of two months after the application of load increment. How many months will be required to reach a settlement of 7.5 cm? What is the settlement in 18 months? The layer has double drainage. [7]

Q.6 (a) A homogeneous clay layer 12 m thick is expected to have an ultimate settlement of 332 mm. After a time span of 3 years, the average settlement was measured to be 152 mm. How much longer will it take for the average settlement to attain 237 mm? [7]

(b) A layer of clay 2 m thick is subjected to a loading of 0.5 kg/cm^2. One year after loading, the average consolidation is 50%. The layer has double drainage, (i) what is the coefficient of consolidation? (ii) If the coefficient permeability is 3 mm/year, what is the settlement after one years, and how much time will the layer take to reach 90% consolidation? [7]

Q.7 (a) Undrained soil sample 30 m thick got 50% consolidation in 20 minutes with drainage allowed at top and bottom in the laboratory. If the clay layer from which the sample was obtained is 3 m thick in field condition, estimate the time it will take to consolidate 50% with (i) double surface drainage (ii) single surface drainage, if in both cases, consolidation pressure is uniform. [7]

(b) In a direct shear test on sand, the normal stress was 2.0 kg/cm^2 and shear stress at failure was 0.8 kN/cm^2. Determine the orientations of the principal planes at failure. [7]

Q.8 (a) A cylindrical sample of soil, having cohesion of 0.8 kg/cm^2 and angle of internal friction of 20^0, is subjected to a cell pressure of 1.0 kg/cm^2. Calculate the maximum deviator stress at which the sample will fail and the angle made by the failure plane with the axis of the sample. [7]

(b) In an unconfined compression test, a sample of clay 100 mm long and 50 mm in diameter fails under a load of 150 N at 10% strain. Calculate the shearing resistance taking into account the effect of change in cross-section of the sample. [7]

Q.9 (a) A shear vane of 7.5 cm diameter and 11.0 cm length was used to measure the shear strength of soft clay. If torque of 600 kg-cm was required to shear the soil, calculate the shear strength. The vane was then rotated rapidly to cause remoulding of the soil. The torque required in the remoulded state was 200 kg-cm. Determine the sensitivity of the soil and offer your comments on the same. [7]

(b) A direct shear test is run on a medium dense sandy silt. At failure, normal stress is 65 kPa and shear stress is 41 kPa. Draw Mohr's circle and determine (i) the angle of shearing resistance, (ii) principal stresses at failure, (iii) orientation of failure plane, principal planes and the plane of maximum shear stress at failure. [7]

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